

# Analog Integrated Circuit Design

Cadence Tools

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## Lab 01

Low Pass Filter Simulation and MOSFET Characteristics

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## 1 Part 1: Low Pass Filter Simulation

### 1.1 Circuit Design

$R = 1\text{ k}\Omega$  and a time constant  $\tau = 0.5\text{ ns}$ .

$$\tau = RC \implies C = \frac{\tau}{R} \quad (1)$$

Substituting the given values:

$$C = \frac{\tau}{R} = \frac{0.5 \times 10^{-9}\text{ s}}{1 \times 10^3\ \Omega} = 0.5 \times 10^{-12}\text{ F} = 0.5\text{ pF} \quad (2)$$

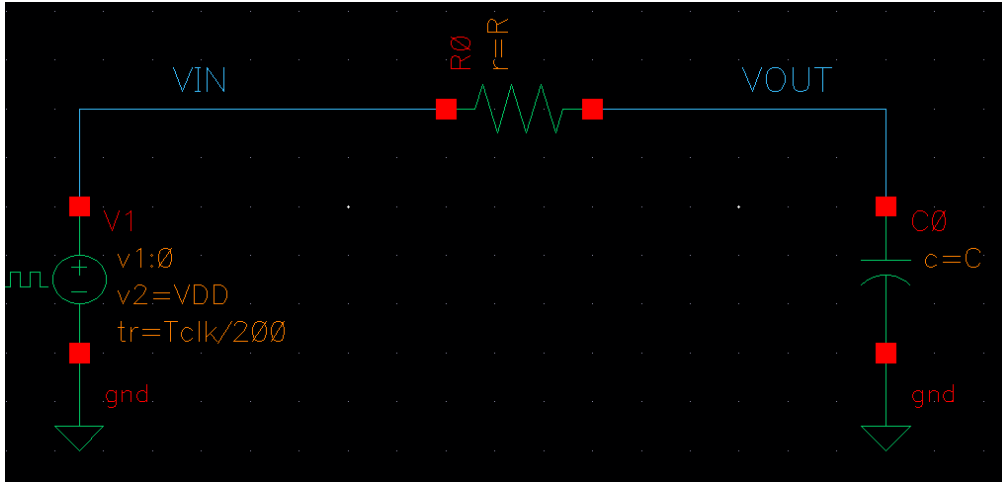


Figure 1: Schematic of the designed RC low pass filter ( $R = 1\text{ k}\Omega$ ,  $C = 0.5\text{ pF}$ ).

The transfer function of this filter is

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{1/sC}{R + 1/sC} = \frac{1}{1 + sRC} = \frac{1}{1 + s\tau} \quad (3)$$

which is a single-pole low pass response with pole at  $s = -1/\tau$ , DC gain of unity (0 dB), and 3 dB bandwidth  $\omega_{3dB} = 1/\tau$ .

### 1.2 Transient Analysis

#### 1.2.1 Input Stimulus

- Period:  $T_{clk} = 52\text{ ns}$
- Pulse width:  $T_{high} = T_{clk}/2 = 26\text{ ns}$
- Rise/Fall time:  $T_{rise} = T_{fall} = T_{clk}/200 = 260\text{ ps}$

|             |            |     |
|-------------|------------|-----|
| Voltage 1   | 0 V        | off |
| Voltage 2   | VDD V      | off |
| Period      | Tclk s     | off |
| Delay time  |            | off |
| Rise time   | Tclk/200 s | off |
| Fall time   | Tclk/200 s | off |
| Pulse width | Tclk/2 s   | off |

(a) Cadence Inputs

#### Random Number Generator

This version of the generator creates a random integer. It can deal with very large integers up to a few thousand digits.

Result

52

|                                                                              |     |
|------------------------------------------------------------------------------|-----|
| Lower Limit                                                                  | 11  |
| Upper Limit                                                                  | 100 |
| <input type="button" value="Generate"/> <input type="button" value="Clear"/> |     |

(b) Randomly-assigned period value ( $T_{clk} = 52\text{ ns}$ )

The maximum simulation time step required by the handout is  $T_{clk}/100$ :

$$\Delta t_{max} = \frac{T_{clk}}{100} = \frac{52 \text{ ns}}{100} = 0.52 \text{ ns} = 520 \text{ ps} \quad (4)$$

$$Rise_{time} = Fall_{time} = \frac{T_{clk}}{200} = 260 \text{ ps} \quad (5)$$

### 1.2.2 Hand Analysis: Step Response

The time-domain output during charging is

$$v_{out}(t) = V_f + (V_i - V_f) e^{-t/\tau} \quad (6)$$

where  $V_i$  is the initial voltage and  $V_f$  is the final (settled) voltage. For a rising edge ( $V_i = 0$ ,  $V_f = V_{step} = 1.8V$ ):

$$v_{out}(t) = V_{step} \left(1 - e^{-t/\tau}\right) \quad (7)$$

and for a falling edge ( $V_i = V_{step} = 1.8V$ ,  $V_f = 0$ ):

$$v_{out}(t) = V_{step} e^{-t/\tau} \quad (8)$$

### 1.2.3 Hand Analysis: Rise and Fall Time (10%–90%)

$v(t)/V_{step} = 1 - e^{-t/\tau}$ . The time  $t_1$  at which the output reaches 10% of its final value satisfies

$$0.1 = 1 - e^{-t_1/\tau} \implies e^{-t_1/\tau} = 0.9 \implies t_1 = -\tau \ln(0.9) = \tau \ln\left(\frac{1}{0.9}\right) \quad (9)$$

The time  $t_2$  at which the output reaches 90% of its final value satisfies

$$0.9 = 1 - e^{-t_2/\tau} \implies e^{-t_2/\tau} = 0.1 \implies t_2 = -\tau \ln(0.1) = \tau \ln(10) \quad (10)$$

The rise time is therefore

$$t_r = t_2 - t_1 = \tau \ln(10) - \tau \ln\left(\frac{1}{0.9}\right) = \tau \left[\ln(10) + \ln(0.9)\right] = \tau \ln(9) \quad (11)$$

By the symmetric exponential decay of the falling edge, the same derivation applies and the fall time equals the rise time:

$$t_f = t_r = \tau \ln(9) \quad (12)$$

Numerically, with  $\tau = 0.5 \text{ ns}$ :

$$t_r = t_f = 0.5 \text{ ns} \times \ln(9) = 0.5 \text{ ns} \times 2.1972 = 1.0986 \text{ ns} \quad (13)$$

### 1.2.4 Transient Simulation Results

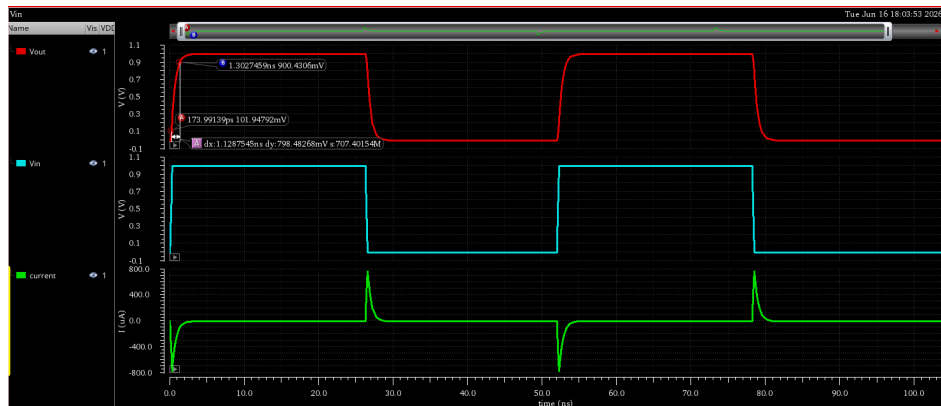


Figure 3: Transient analysis results

### 1.2.5 Comparison Table: Simulation vs. Analytical (Rise/Fall Time)

|               |           |        |
|---------------|-----------|--------|
| ITl:LAB01_1:1 | rise_time | 1.124n |
| ITl:LAB01_1:1 | fall_time | 1.125n |

Figure 4: Rise and fall time

Table 1: Comparison between hand-calculated and simulated rise/fall time.

| Quantity                  | Analytical (hand calc.) | Cadence Simulation | % Error |
|---------------------------|-------------------------|--------------------|---------|
| Rise time $t_r$ (10%–90%) | 1.0986 ns               | 1.124 ns           | 2.312   |
| Fall time $t_f$ (10%–90%) | 1.0986 ns               | 1.125 ns           | 2.4     |

### 1.3 Parametric Sweep: $R = 1 : 1 : 5 \text{ k}\Omega$

$R$  is swept from  $1 \text{ k}\Omega$  to  $5 \text{ k}\Omega$  in steps of  $1 \text{ k}\Omega$ .

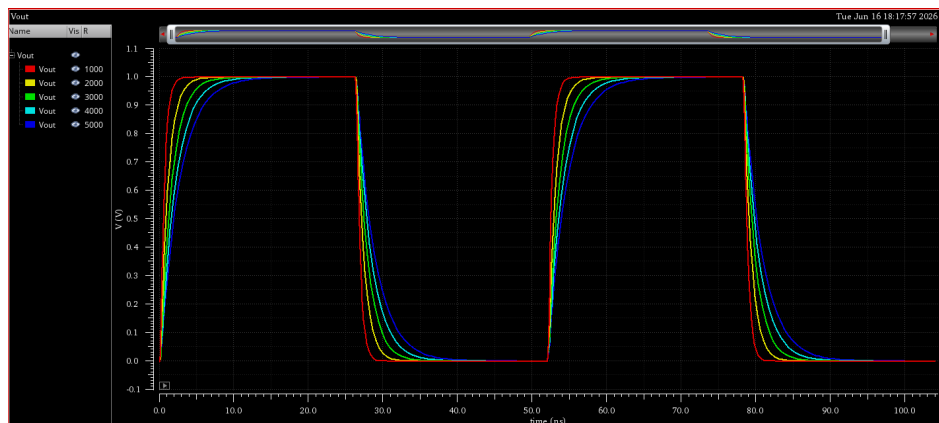


Figure 5: Vout over different R

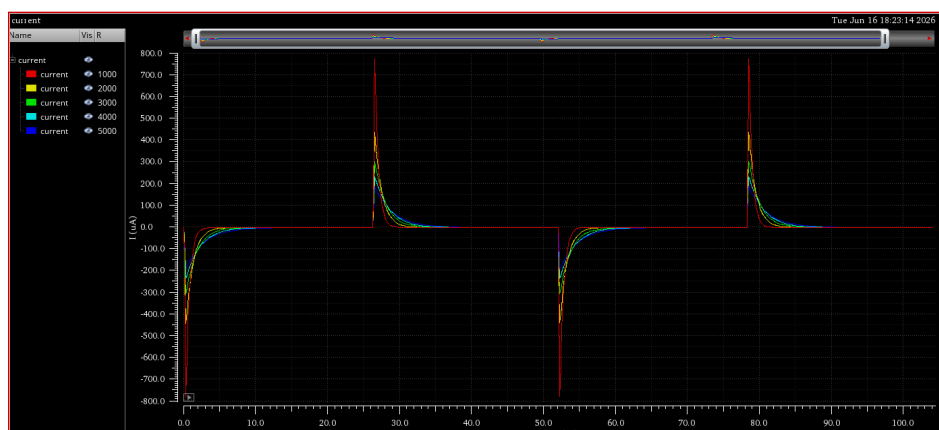
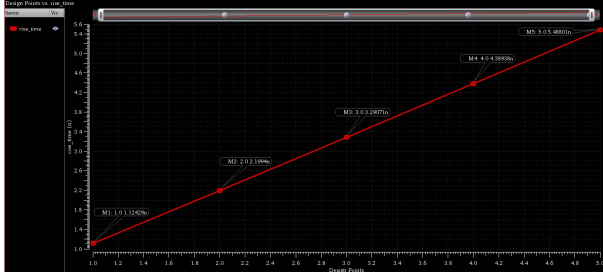
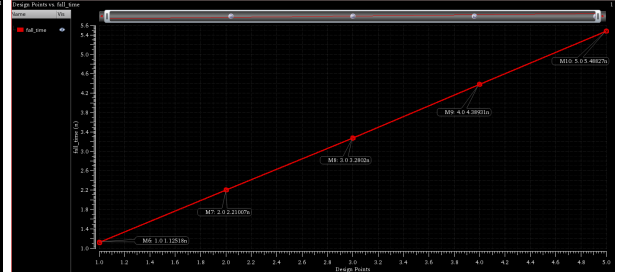


Figure 6: Current over different R



(a) Rise time over R sweep



(b) Fall time over R sweep

### Comments

As  $R$  increases from  $1\text{ k}\Omega$  to  $5\text{ k}\Omega$  with  $C$  fixed, the time constant  $\tau = RC$  increases proportionally (linearly) with  $R$ . Consequently:

- The output waveform charges and discharges more slowly – the exponential curves become visibly "lazier," with more rounded corners and a longer approach to the final value.
- The rise time and fall time both increase linearly with  $R$  (since  $t_r = t_f = RC \ln 9$ ), exactly as tabulated in Table 2: at  $R = 5\text{ k}\Omega$  the rise time is  $5\times$  that at  $R = 1\text{ k}\Omega$ .
- The 3 dB bandwidth decreases inversely with  $R$  (i.e.  $f_{3dB} \propto 1/R$ ): at  $R = 5\text{ k}\Omega$ , the bandwidth is  $1/5$  of that at  $R = 1\text{ k}\Omega$ .
- This demonstrates the fundamental RC low pass filter trade-off: increasing  $R$  (or  $C$ ) makes the filter "slower" in the time domain (longer rise/fall times) but more selective in the frequency domain (lower bandwidth, more aggressive filtering of high frequencies).

For each value of  $R$ , the new time constant is  $\tau = RC$ , and the analytical rise/fall time and 3 dB bandwidth scale linearly and inversely with  $R$  respectively:

$$\tau(R) = RC, \quad t_r(R) = t_f(R) = \tau(R) \ln 9, \quad f_{3dB}(R) = \frac{1}{2\pi\tau(R)} = \frac{1}{2\pi RC} \quad (14)$$

Table 2: Hand-calculated  $\tau$ , rise/fall time, and bandwidth vs. swept  $R$  ( $C = 0.5\text{ pF}$  fixed).

| $R$ (k $\Omega$ ) | $\tau = RC$ (ns) | $t_r = t_f = \tau \ln 9$ (ns) | $f_{3dB} = \frac{1}{2\pi\tau}$ (MHz) |
|-------------------|------------------|-------------------------------|--------------------------------------|
| 1                 | 0.500            | 1.0986                        | 318.31                               |
| 2                 | 1.000            | 2.1972                        | 159.15                               |
| 3                 | 1.500            | 3.2958                        | 106.10                               |
| 4                 | 2.000            | 4.3944                        | 79.58                                |
| 5                 | 2.500            | 5.4931                        | 63.66                                |

| Point            | Test          | Output    | Nominal | Spec | Weight | Pass/Fail |
|------------------|---------------|-----------|---------|------|--------|-----------|
| Parameters: R=1k |               |           |         |      |        |           |
| 1                | ITi:LAB01_1:1 | rise_time | 1.124n  |      |        |           |
| 1                | ITi:LAB01_1:1 | fall_time | 1.125n  |      |        |           |
| 1                | ITi:LAB01_1:1 | Vout      |         |      |        |           |
| 1                | ITi:LAB01_1:1 | Vin       |         |      |        |           |
| 1                | ITi:LAB01_1:1 | current   |         |      |        |           |
| Parameters: R=2k |               |           |         |      |        |           |
| 2                | ITi:LAB01_1:1 | rise_time | 2.199n  |      |        |           |
| 2                | ITi:LAB01_1:1 | fall_time | 2.21n   |      |        |           |
| 2                | ITi:LAB01_1:1 | Vout      |         |      |        |           |
| 2                | ITi:LAB01_1:1 | Vin       |         |      |        |           |
| 2                | ITi:LAB01_1:1 | current   |         |      |        |           |
| Parameters: R=3k |               |           |         |      |        |           |
| 3                | ITi:LAB01_1:1 | rise_time | 3.291n  |      |        |           |
| 3                | ITi:LAB01_1:1 | fall_time | 3.28n   |      |        |           |
| 3                | ITi:LAB01_1:1 | Vout      |         |      |        |           |
| 3                | ITi:LAB01_1:1 | Vin       |         |      |        |           |
| 3                | ITi:LAB01_1:1 | current   |         |      |        |           |
| Parameters: R=4k |               |           |         |      |        |           |
| 4                | ITi:LAB01_1:1 | rise_time | 4.389n  |      |        |           |
| 4                | ITi:LAB01_1:1 | fall_time | 4.389n  |      |        |           |
| 4                | ITi:LAB01_1:1 | Vout      |         |      |        |           |
| 4                | ITi:LAB01_1:1 | Vin       |         |      |        |           |
| 4                | ITi:LAB01_1:1 | current   |         |      |        |           |
| Parameters: R=5k |               |           |         |      |        |           |
| 5                | ITi:LAB01_1:1 | rise_time | 5.488n  |      |        |           |
| 5                | ITi:LAB01_1:1 | fall_time | 5.488n  |      |        |           |
| 5                | ITi:LAB01_1:1 | Vout      |         |      |        |           |

Figure 8: Sweep Results

## 1.4 AC Analysis

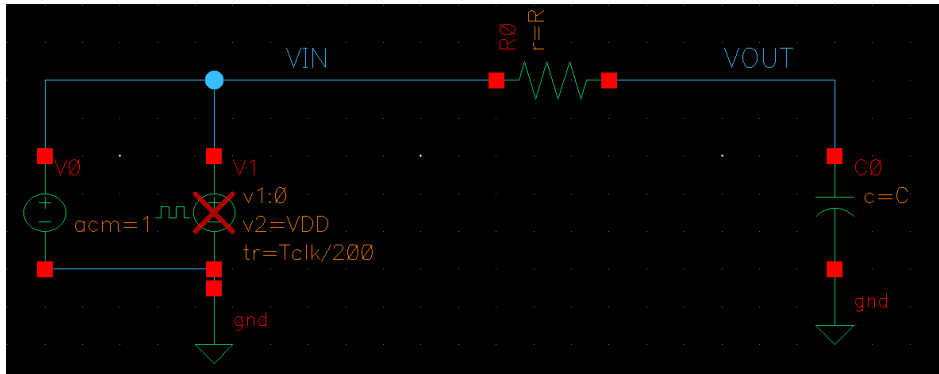


Figure 9: LPF schematic in AC analysis

### 1.4.1 Hand Analysis: Transfer Function, DC Gain, and Bandwidth

Starting from the transfer function derived earlier,

$$H(s) = \frac{1}{1 + sRC} = \frac{1}{1 + s\tau} \quad (15)$$

substituting  $s = j\omega$ :

$$H(j\omega) = \frac{1}{1 + j\omega\tau} \quad (16)$$

The magnitude response is

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega\tau)^2}} \quad (17)$$

and the phase response is

$$\angle H(j\omega) = -\tan^{-1}(\omega\tau) \quad (18)$$

**DC Gain:** Evaluating at  $\omega = 0$ :

$$|H(j0)| = \frac{1}{\sqrt{1 + 0}} = 1 \implies \text{DC Gain (dB)} = 20 \log_{10}(1) = 0 \text{ dB} \quad (19)$$

**3 dB Bandwidth:**

$$\frac{1}{\sqrt{1 + (\omega_{3dB}\tau)^2}} = \frac{1}{\sqrt{2}} \Rightarrow 1 + (\omega_{3dB}\tau)^2 = 2 \Rightarrow \omega_{3dB}\tau = 1 \Rightarrow \omega_{3dB} = \frac{1}{\tau} \quad (20)$$

Converting to Hz:

$$f_{3dB} = \frac{\omega_{3dB}}{2\pi} = \frac{1}{2\pi\tau} = \frac{1}{2\pi RC} \quad (21)$$

Numerically, with  $\tau = 0.5 \text{ ns}$  ( $R = 1 \text{ k}\Omega$ ,  $C = 0.5 \text{ pF}$ ):

$$f_{3dB} = \frac{1}{2\pi \times 0.5 \times 10^{-9}} = \frac{1}{3.1416 \times 10^{-9}} = 3.1831 \times 10^8 \text{ Hz} \approx 318.31 \text{ MHz} \quad (22)$$

**Phase at the 3 dB frequency:**

$$\angle H(j\omega_{3dB}) = -\tan^{-1}(\omega_{3dB}\tau) = -\tan^{-1}(1) = -45^\circ \quad (23)$$

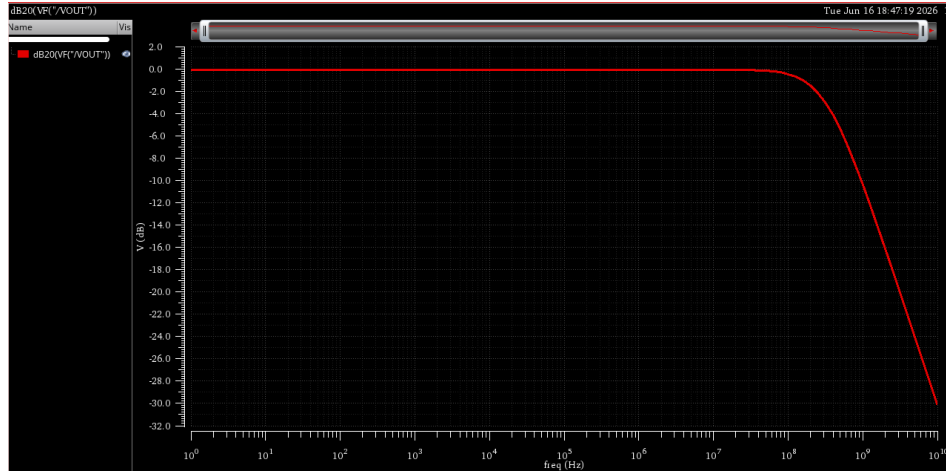
**1.4.2 AC Simulation Results**

Figure 10: AC analysis: magnitude response in dB.

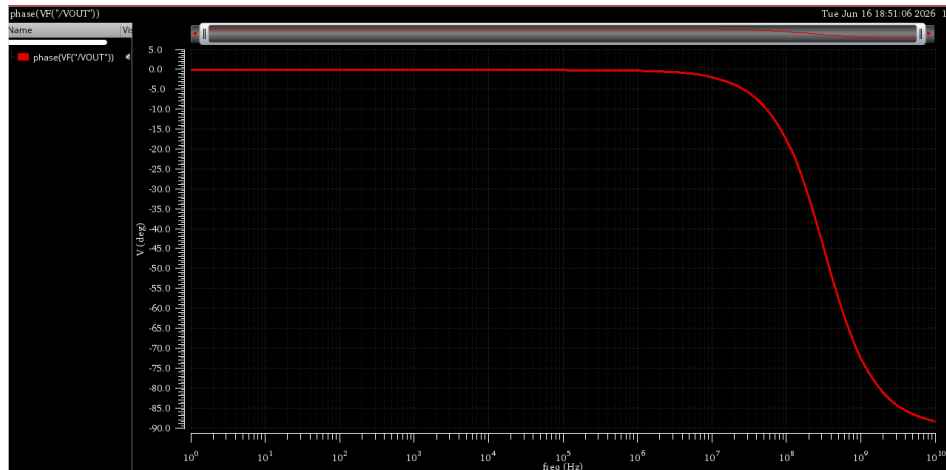


Figure 11: AC analysis: phase response

|               |                                 |        |
|---------------|---------------------------------|--------|
| IT1:LAB01_1:1 | bandwidth(VF("/VOUT")) 3 "lo... | 317.7M |
| IT1:LAB01_1:1 | ymin(mag(VF("/VOUT"))) 1        | 1      |

Figure 12: Result for the 3 dB bandwidth



### 1.4.3 Comparison Table: Simulation vs. Analytical (AC)

Table 3: Comparison between hand-calculated and simulated DC gain and bandwidth.

| Quantity                 | Analytical (hand calc.) | Cadence Simulation | % Error |
|--------------------------|-------------------------|--------------------|---------|
| DC Gain                  | 0 dB (unity)            | 0 dB               | 0       |
| 3 dB Bandwidth $f_{3dB}$ | 318.31 MHz              | 317.7 MHz          | 0.1916  |

### 1.5 Parametric AC Sweep: $R = 1, 10, 100, 1000 \text{ k}\Omega$

Here  $R$  is swept over four decades while  $C$  remains fixed at  $0.5 \text{ pF}$ , so  $\tau = RC$  also spans four decades, and consequently  $f_{3dB} = 1/(2\pi RC)$  spans four decades (inversely).

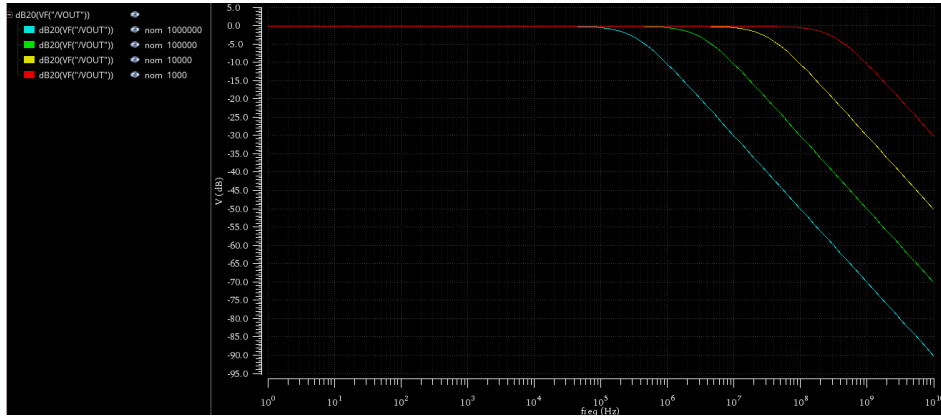


Figure 13: Output voltage magnitude in dB across frequency

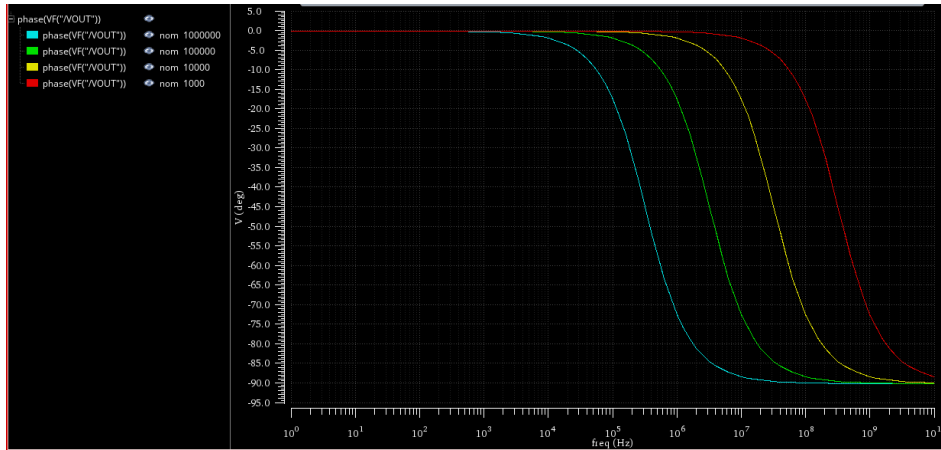


Figure 14: Output voltage phase across frequency

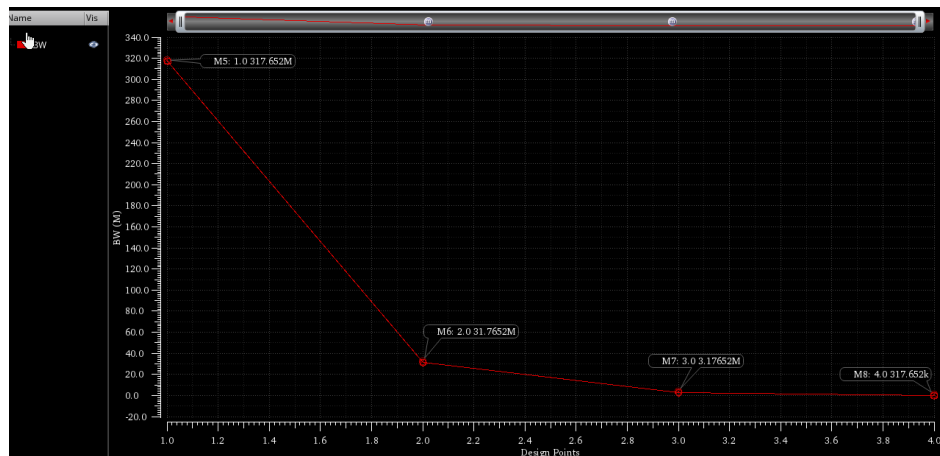


Figure 15: 3dB BW across design points in linear scale



Figure 16: 3dB BW across design points in log scale

| Parameters: R=1k   |               |                        |  |        |  |
|--------------------|---------------|------------------------|--|--------|--|
| 1                  | ITi:LAB01_1:1 | dB20(VF("/VOUT"))      |  |        |  |
| 1                  | ITi:LAB01_1:1 | mag(VF("/VOUT"))       |  |        |  |
| 1                  | ITi:LAB01_1:1 | VF("/VOUT")            |  |        |  |
| 1                  | ITi:LAB01_1:1 | phase(VF("/VOUT"))     |  |        |  |
| 1                  | ITi:LAB01_1:1 | BW                     |  | 317.7M |  |
| 1                  | ITi:LAB01_1:1 | ymax(mag(VF("/VOUT"))) |  | 1      |  |
| Parameters: R=10k  |               |                        |  |        |  |
| 2                  | ITi:LAB01_1:1 | dB20(VF("/VOUT"))      |  |        |  |
| 2                  | ITi:LAB01_1:1 | mag(VF("/VOUT"))       |  |        |  |
| 2                  | ITi:LAB01_1:1 | VF("/VOUT")            |  |        |  |
| 2                  | ITi:LAB01_1:1 | phase(VF("/VOUT"))     |  |        |  |
| 2                  | ITi:LAB01_1:1 | BW                     |  | 31.77M |  |
| 2                  | ITi:LAB01_1:1 | ymax(mag(VF("/VOUT"))) |  | 1      |  |
| Parameters: R=100k |               |                        |  |        |  |
| 3                  | ITi:LAB01_1:1 | dB20(VF("/VOUT"))      |  |        |  |
| 3                  | ITi:LAB01_1:1 | mag(VF("/VOUT"))       |  |        |  |
| 3                  | ITi:LAB01_1:1 | VF("/VOUT")            |  |        |  |
| 3                  | ITi:LAB01_1:1 | phase(VF("/VOUT"))     |  |        |  |
| 3                  | ITi:LAB01_1:1 | BW                     |  | 3.177M |  |
| 3                  | ITi:LAB01_1:1 | ymax(mag(VF("/VOUT"))) |  | 1      |  |
| Parameters: R=1M   |               |                        |  |        |  |
| 4                  | ITi:LAB01_1:1 | dB20(VF("/VOUT"))      |  |        |  |
| 4                  | ITi:LAB01_1:1 | mag(VF("/VOUT"))       |  |        |  |
| 4                  | ITi:LAB01_1:1 | VF("/VOUT")            |  |        |  |
| 4                  | ITi:LAB01_1:1 | phase(VF("/VOUT"))     |  |        |  |
| 4                  | ITi:LAB01_1:1 | BW                     |  | 317.7k |  |
| 4                  | ITi:LAB01_1:1 | ymax(mag(VF("/VOUT"))) |  | 1      |  |

Figure 17: Summary of parametric sweep from cadence

## Comments

As  $R$  increases by a factor of 10 at each step ( $1\text{ k}\Omega \rightarrow 10\text{ k}\Omega \rightarrow 100\text{ k}\Omega \rightarrow 1000\text{ k}\Omega$ ), the time constant  $\tau = RC$  also increases by a factor of 10 at each step, and so the 3 dB bandwidth  $f_{3dB} = 1/(2\pi RC)$  decreases by exactly a factor of 10 at each step – this is why the four overlaid Bode magnitude curves in Fig. 13 appear as identical-shape curves shifted by exactly one decade along the (logarithmic) frequency axis relative to each other. The DC gain remains exactly 0 dB (unity) for all four cases, since DC gain of a simple RC voltage divider with no loading is independent of the absolute value of  $R$  (it only depends on the ratio of impedances, and at DC the capacitor impedance is infinite regardless of  $R$ , so  $V_{out} = V_{in}$  in all four cases). This sweep nicely demonstrates that  $R$  (or equivalently  $C$ ) is the single "knob" that sets the corner frequency of a passive RC low pass filter, while the passband (DC) gain is fixed by the topology and unaffected by the time-constant value.

## 1.6 [Optional] Pole-Zero Analysis

### 1.6.1 Hand Analysis

The transfer function  $H(s) = \frac{1}{1 + s\tau}$  can be rewritten in pole form as

$$H(s) = \frac{1}{\tau} \cdot \frac{1}{s + \frac{1}{\tau}} \times \tau = \frac{1/\tau}{s + 1/\tau} \quad (24)$$

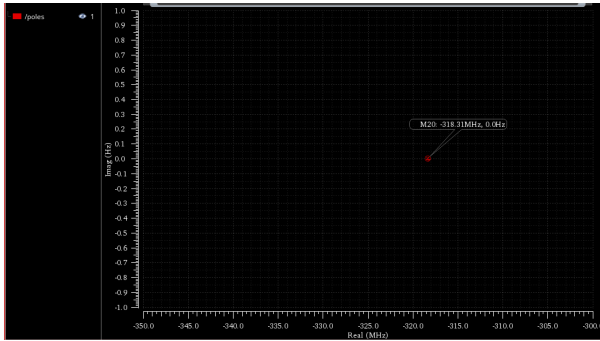
so the system has a single real pole located at

$$s_p = -\frac{1}{\tau} = -\frac{1}{RC} \quad (25)$$

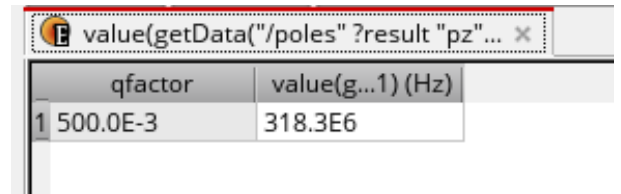
and no finite zeros. Numerically:

$$s_p = -\frac{1}{0.5 \times 10^{-9}} = -2 \times 10^9 \text{ rad/s} \quad (26)$$

$$f_p = \frac{2 \times 10^9}{2\pi} = 3.1831 \times 10^8 \text{ Hz} = 318.31 \text{ MHz} \quad (27)$$



(a) pole over real and imag axis



(b) calculated pole

Table 4: Comparison between hand-calculated pole frequency, AC-analysis bandwidth, and pole-zero analysis result.

| Quantity                                  | Hand Calculation | Cadence Result |
|-------------------------------------------|------------------|----------------|
| Pole frequency $f_p$ (Pole-Zero analysis) | 318.31 MHz       | 318.3 MHz      |
| 3 dB Bandwidth (AC analysis)              | 318.31 MHz       | 317.7 MHz      |

## 2 Part 2: MOSFET Characteristics

### 2.1 Technology and Device Setup

This part uses a **180 nm** CMOS technology with  $V_{DD} = 1.8\text{ V}$ .

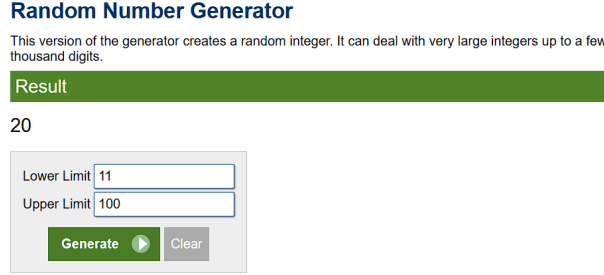


Figure 19: Randomly-assigned aspect ratio value used for this report ( $W/L = 20$ )

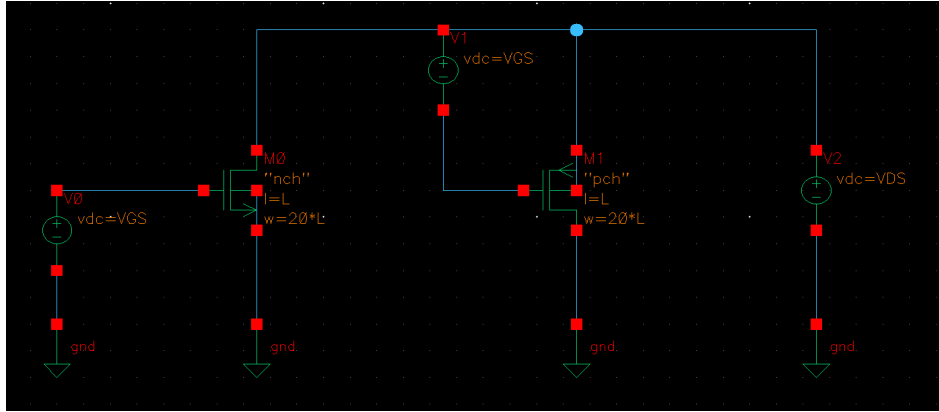


Figure 20: Schematic design for NMOS and PMOS

Table 5: Device geometries used in this report.

| Device        | $L$             | $W$              | $W/L$ |
|---------------|-----------------|------------------|-------|
| Short channel | 200 nm          | 4 $\mu\text{m}$  | 20    |
| Long channel  | 2 $\mu\text{m}$ | 40 $\mu\text{m}$ | 20    |

### 2.2 $I_D$ vs. $V_{GS}$

#### 2.2.1 Simulation Setup

$V_{GS} = 0 : 10\text{mV} : V_{DD}$ ,  $V_{DS} = V_{DD} = 1.8\text{ V}$  for all points. The short-channel and long-channel devices are simulated together using a parametric sweep over  $L \in \{200\text{nm}, 2\mu\}$  with  $W$  defined as the expression  $W = 20L$ .

#### 2.2.2 Hand Analysis: NMOS Square-Law Equations

For an NMOS transistor with  $V_{DS} = V_{DD}$  and  $V_{GS}$  swept from 0 to  $V_{DD}$ , the transistor transitions through three regions:

**Cutoff** ( $V_{GS} < V_{TH}$ ):

$$I_D = 0 \quad (28)$$

**Saturation** ( $V_{GS} \geq V_{TH}$  and  $V_{DS} \geq V_{ov}$ . since  $V_{DS} = V_{DD} = 1.8\text{V}$  is fixed and large, the device remains in saturation for essentially the entire useful range of  $V_{GS}$  above threshold (as long as  $V_{DD} \geq V_{GS} - V_{TH}$ , which holds since  $V_{GS} \leq V_{DD}$  and  $V_{TH} > 0$ ):

$$I_D = \frac{1}{2} k'_n \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS}) \quad (29)$$

This is a **quadratic (square-law)** relationship between  $I_D$  and the overdrive voltage ( $V_{GS} - V_{TH}$ ).

### 2.2.3 Simulation Results

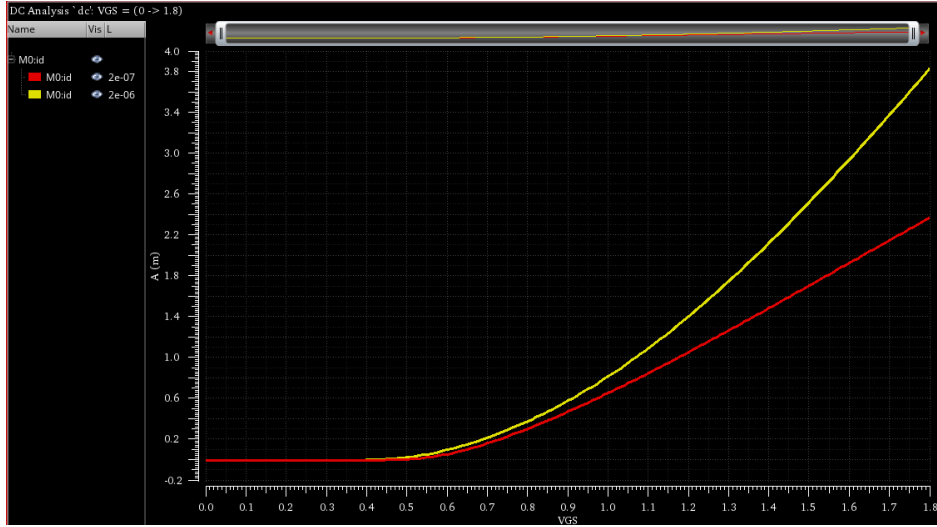


Figure 21:  $I_D$ – $V_{GS}$  characteristics for NMOS, short-channel(Red) and long-channel(Yellow).

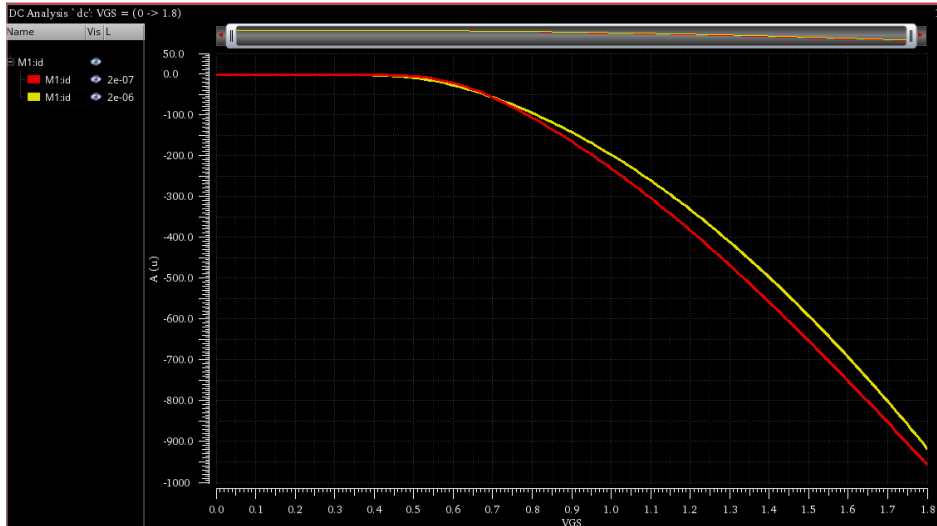
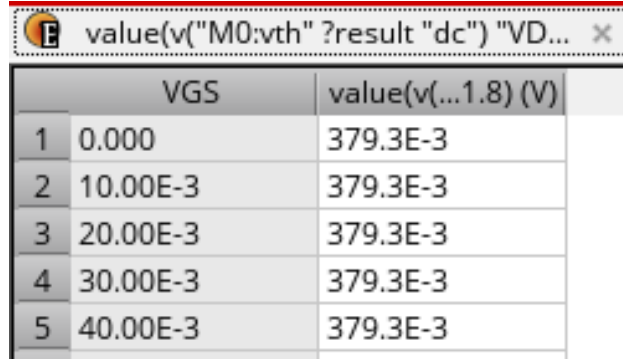


Figure 22:  $I_D$ – $V_{GS}$  characteristics for PMOS, short-channel(Red) and long-channel(Yellow).



|   | VGS      | value(v(...1.8) (V)) |
|---|----------|----------------------|
| 1 | 0.000    | 379.3E-3             |
| 2 | 10.00E-3 | 379.3E-3             |
| 3 | 20.00E-3 | 379.3E-3             |
| 4 | 30.00E-3 | 379.3E-3             |
| 5 | 40.00E-3 | 379.3E-3             |

Figure 23: Extracted threshold voltage for NMOS

### Comments

#### 1) Short-channel vs. long-channel – which has higher current, and why?

The Long Channel Device has higher current, due to the lower electric field in the channel allowing carriers to move more freely, While in short-channel the carriers' velocity saturates limiting the current. The effect is more noticeable in NMOS devices due to the higher carrier mobility.

#### 2) Is the relation linear or quadratic, and why?

- Long channel in NMOS: The relationship is quadratic. Since the electric field is relatively low, carriers do not reach velocity saturation, and the device follows the ideal square-law mode  $I_D \propto (V_{GS} - V_{TH})^2$ .
- Short channel in PMOS: The relationship becomes linear at higher  $V_{GS}$  values. Because the carriers reach  $v_{sat}$  quickly, the current depends on the saturation velocity rather than mobility, changing the proportionality to  $I_D \propto (V_{GS} - V_{TH})$ .
- In PMOS transistors, short-channel effects such as channel-length modulation and threshold voltage roll-off can cause a reduction in the effective channel length, which increases in the drain current and a small deviation from the ideal quadratic relationship between the drain current and channel length. While it is true that the mobility of holes is lower than that of electrons in NMOS transistors, short-channel effects can still occur in PMOS transistors.

#### 3) NMOS vs. PMOS – which has higher current, and what is the ratio at $V_{GS} = V_{DD}$ ?

NMOS has higher current than PMOS for the same  $W/L$ , because electron mobility  $\mu_n$  is intrinsically higher than hole mobility  $\mu_p$  in silicon (electrons experience less scattering than holes for comparable doping/field conditions), making  $k'_n = \mu_n C_{ox} > k'_p = \mu_p C_{ox}$ . 4) what is the ratio at  $V_{GS} = V_{DD}$ ?

- For the short channel ( $L = 0.2\mu\text{m}$ ):

$$ratio = \frac{I_D(\text{NMOS})}{I_D(\text{PMOS})} \Big|_{V_{GS}=V_{DD}} = \frac{2.35}{0.96} \approx 2.45 \quad (30)$$

- For the long channel ( $L = 2\mu\text{m}$ ):

$$ratio = \frac{I_D(\text{NMOS})}{I_D(\text{PMOS})} \Big|_{V_{GS}=V_{DD}} = \frac{3.85}{0.91} \approx 4.2 \quad (31)$$

#### Which is more affected by short-channel effects?

Generally, NMOS is more affected by short-channel effects than PMOS because electrons have higher intrinsic mobility making them more prone to velocity saturation at a given lateral field (they reach  $v_{sat}$  at a lower field than holes)

### 2.3 $g_m$ vs. $V_{GS}$

#### 2.3.1 Hand Analysis

$$g_m = \frac{\partial I_D}{\partial V_{GS}} \quad (32)$$

For a **long-channel** device operating in saturation (ideal square law), we have:

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH}) \quad (33)$$

Thus, for the long-channel transistor,  $g_m$  increases linearly with the overdrive voltage.

For a **short-channel** device under high electric fields, velocity saturation occurs, and the current relationship becomes:

$$I_D \approx WC_{ox}v_{sat}(V_{GS} - V_{TH}) \quad (34)$$

Taking the derivative, the transconductance becomes:

$$g_m \approx WC_{ox}v_{sat} \quad (35)$$

which is a constant value independent of  $V_{GS}$ .

### 2.3.2 Simulation Results

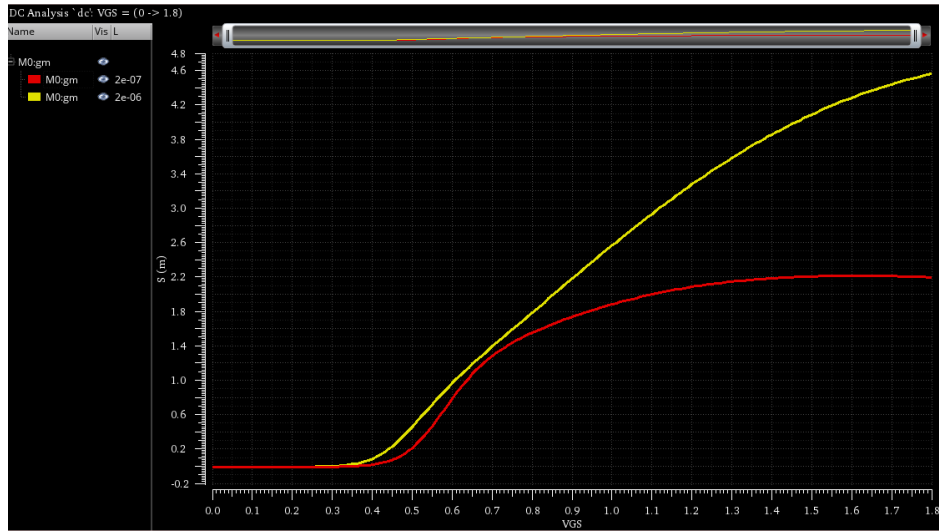


Figure 24: Transconductance ( $g_m$ ) vs  $V_{GS}$  for NMOS short(Red) and long channel(yellow).

### Comments

#### 1) Does $g_m$ increase linearly? Why?

For the long-channel device,  $g_m$  increases linearly because the derivative of the quadratic drain current with respect to  $V_{GS}$  yields a linear function of  $(V_{GS} - V_{TH})$ . For short-channel NMOS does not increase linearly with  $V_{GS}$ , due to the influence of short-channel effects.

#### 2) Does $g_m$ saturate? Why?

For the short-channel device,  $g_m$  saturates and flattens out at higher  $V_{GS}$ . This happens because the carriers in the short channel reach their saturation velocity ( $v_{sat}$ ) due to the high lateral electric field. As a result, the drain current becomes linearly dependent on  $V_{GS}$ , and its derivative (which is  $g_m$ ) becomes a constant.

## 2.4 $I_D$ vs. $V_{DS}$

### 2.4.1 Simulation Setup

The  $I_D - V_{DS}$  characteristics are obtained using a nested sweep. A primary DC sweep is performed on  $V_{DS}$  from 0 V to  $V_{DD}$  with a 10 mV step, while a secondary parametric sweep is applied to  $V_{GS}$  from 0 V to  $V_{DD}$  with a step of 0.2 V.

### 2.4.2 Simulation Results

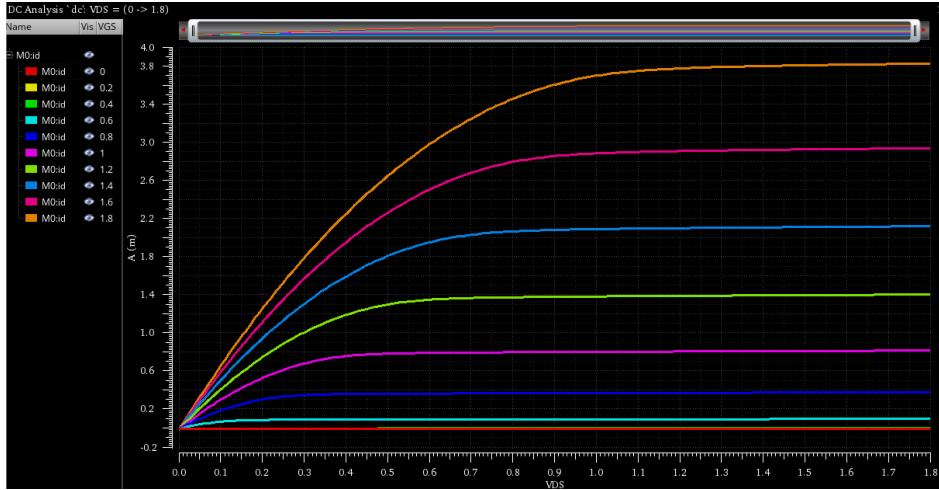


Figure 25: Long Channel NMOS( $L = 2 \mu\text{m}$ ) drain current vs VDS

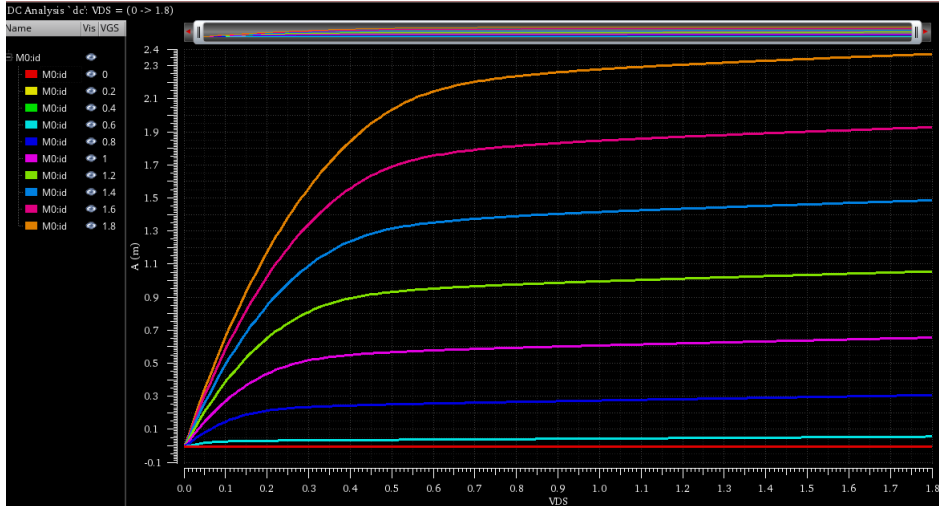


Figure 26: Short Channel NMOS( $L = 200 \text{ nm}$ ) drain current vs VDS

#### Comments

##### 1) Which one has higher current? Why?

For the same aspect ratio ( $W/L = 20$ ), the long-channel device allows a relatively higher normalized current at large overdrive voltages because the short-channel device suffers heavily from velocity saturation and mobility degradation, which limits the maximum current it can provide.

##### 2) Which one has a higher slope in the saturation region? Why?

The short-channel device has a significantly higher slope in the saturation region. This non-zero slope indicates a lower output resistance ( $r_o$ ). This is caused primarily by severe Channel Length Modulation (CLM). In a short channel, the drain depletion region occupies a much larger fraction of the physical gate length, thereby heavily modulating the effective channel length as  $V_{DS}$  increases.

## 2.5 [Optional] $g_m$ and $r_o$ in Triode and Saturation

### 2.5.1 Hand Analysis

The output resistance  $r_o$  is defined as:

$$r_o = \left( \frac{\partial I_D}{\partial V_{DS}} \right)^{-1} \quad (36)$$



In the **triode region**,  $I_D$  is strongly dependent on  $V_{DS}$ :

$$I_D \approx \mu_n C_{ox} \frac{W}{L} \left( (V_{GS} - V_{TH}) V_{DS} - \frac{V_{DS}^2}{2} \right) \quad (37)$$

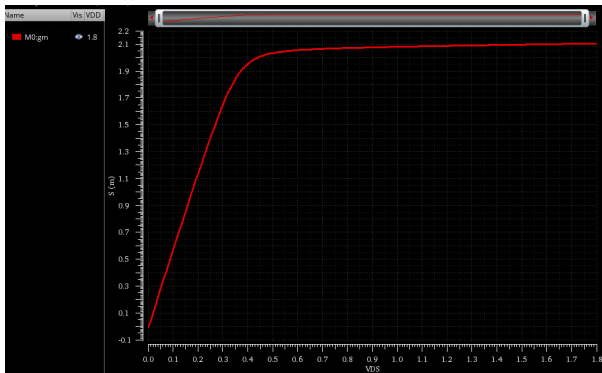
Taking the derivative for transconductance gives  $g_m \approx \mu_n C_{ox} \frac{W}{L} V_{DS}$ , meaning  $g_m$  increases linearly with  $V_{DS}$ . The output resistance  $r_o$  is relatively low.

In the **saturation region**, the channel pinches off and  $I_D$  is ideally independent of  $V_{DS}$  (ignoring CLM):

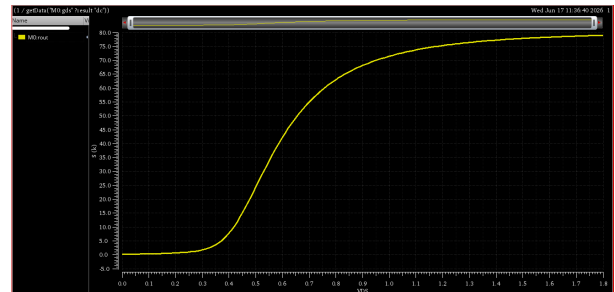
$$r_o \approx \frac{1}{\lambda I_D} \quad (38)$$

Here,  $g_m$  saturates to a constant value dependent on  $V_{GS}$ , and  $r_o$  becomes large.

### 2.5.2 Simulation Results



(a) Transconductance ( $g_m$ ) vs  $V_{DS}$



(b) Output Resistance ( $r_o$ ) vs  $V_{DS}$

Figure 27: Small-signal parameters behavior across operating regions.

### Comments

#### 1) Variation of $g_m$ vs $V_{DS}$ :

- **In the first part of the curve, is the relation linear? Why?** Yes. In the deep triode region,  $I_D \propto V_{DS}$ , making the derivative  $g_m$  linearly dependent on  $V_{DS}$ .
- **Does  $g_m$  saturate? Why?** Yes. As the device enters saturation ( $V_{DS} \geq V_{GS} - V_{TH}$ ), the channel pinches off, and  $I_D$  becomes largely independent of  $V_{DS}$ . Consequently,  $g_m$  stabilizes to a constant value.
- **Where do you want to operate for analog amplifier applications? Why?** In the saturation region. Analog amplifiers require a constant, stable  $g_m$  that does not fluctuate with output voltage swings to ensure linear voltage amplification without severe harmonic distortion.

#### 2) Variation of $r_o$ vs $V_{DS}$ :

- **Does  $r_o$  saturate just after the transistor enters saturation similar to  $g_m$ ?** No, it does not instantly peak or stabilize. The transition between triode and deep saturation is gradual, and  $r_o$  continuously changes as the drain depletion region expands into the channel.
- **Does  $r_o$  increase if the transistor is biased more into saturation?** Yes. Operating deeper into the saturation region (higher  $V_{DS}$ ) generally improves the output resistance before reaching breakdown.
- **Should we operate the transistor at the edge of saturation?** No. At the exact edge of saturation,  $r_o$  is still relatively low and rapidly changing, heavily degrading the intrinsic voltage gain.
- **Where do you want to operate for analog amplifier applications? Why?** We operate deep in the saturation region (with an adequate margin above  $V_{ov}$ ). The intrinsic gain of the transistor is  $A_v = g_m r_o$ . Operating deep in saturation ensures a high output resistance, thereby maximizing the voltage gain.